

This document is an outline which can be used to assist in setting up a HamVOIP Allstar image on SHARI Allstar devices equipped with the K4H818 radio module. Kits 4 Hams models using the new K4H818 radio module are known as SHARI RevK models.

When performing a HamVOIP setup, some of the advanced setup steps like obtaining the K4H818 utilities, installing Supermon, etc are accomplished using commands typed into the Terminal. The required entries to the terminal line are contained in this document as bold text to assist with this by avoiding typing in long complicated commands, This bold text can be copied (highlight then right mouse click) and then pasted to the terminal line (left click to change the focus and then a right click to paste).

To Begin :

- Download and install MobaXterm (an SSH program) [MobaXterm Xserver with SSH, telnet, RDP, VNC and X11 - Download](#)
- Install the latest Raspberry Pi Imager on your computer [Raspberry Pi software – Raspberry Pi](#)
- Download and save the HamVOIP image [Latest HamVOIP Allstar Image](#)
- (Option) Download the K4H818_update info PDF [K4H818-update app information](#)
- (Option) Download the K4H818_prog info PDF [K4H818-prog app information](#)

Write the HamVOIP image to your microSD card using Raspberry Pi Imager:

- Plug your microSD card into your computer using USB adapter or a uSD to SD adapter
- Open Raspberry Pi Imager on your computer
- Select your Raspberry Pi device by scrolling all the way to the bottom and select *No Filtering*
- Click on **Next**
- Scroll all the way to the bottom and select *Use Custom*
- Browse to the HamVOIP image you saved and left click to select it. Click on Open
- Click on **Next**
- Select your storage device (USB Mass Storage Device). Left click on it to highlight it
- Click on **Next**
- Click on **WRITE**
- Click on **I UNDERSTAND, ERASE AND WRITE**
- Wait for the microSD card writing operation to complete and click on **FINISH**

Configure the Node:

- Remove the microSD card and insert in your node's Pi microSD slot (end opposite the USB and ethernet connectors)
- Plug in the ethernet cable to the nodes Pi. The Pi and your computer must be connected to the same local network — like the same router or Wi-Fi — so they can see each other.
- Turn on an HT and set it to the same RF frequency and CTCSS (PL) as is stored in the K4H818

Hint: For the SHARI RevK, this varies as follows:

- For a new SHARI RevK purchased as a kit it is 435.000 MHz with a 100.0 Hz PL
 - For an assembled and tested unit on 2 meters, it is 146.540 MHz with a 100 Hz PL
 - For an assembled and tested unit on 1.25 meters, it is 223.400 MHz with a 100 Hz PL
 - For an assembled and tested unit on 70 cm, it is 446.100 MHz with a 100 Hz PL
- Apply power to the node and listen for an announced IP address (Ignore the PL88.5 announcement)
Hint: You can also log into your router and look for the hostname 'alarmpi' in the Clients list or use your keypad of your radio to send DTMF of *A1 to have the IP address announced again
 - Wait for the blinking green HB LED on the SHARI board to begin blinking (not the green LED on the Pi)

Log in:

- Start the MobaXterm SSH program
- Click on Session icon then click on SSH icon
- Enter the **IP address** into the Remote Host entry input box
- Check *Specify username* and enter **root** for username in the box following the checkmark
- Enter 222 for Port
- Click on OK
- Accept any warning
- Enter **root** for password when prompted

Hint: In Linux password entries are not shown as you type. Just type it in blindly and press <enter>

Updates:

- Click on (accept) Yes to retrieve the latest system updates
- Press <enter> to accept OK to reboot
- The Pi will start to reboot so your MobaXterm session will stop. Wait for the IP announcement on your HT and the blinking green HB LED
- Make sure focus is on MobaXterm and press R to restart the Mobaxterm session
- Press Tab to move to <No> and press <enter> to decline retrieving the latest update again

Set up your node:

- Press <enter> to accept <Yes> to run first setup
- Enter your desired password that you will use to log into your HamVOIP

Hint: Your HamVOIP login user name will be **root** and the password will be **what you just entered**. Save them as the password is not recoverable if you forget it
- Re-enter for verification
- Select No for “Is this a private node”
- Enter your node number (as issued by allstarlink.org)
- Select Yes to setup your node configuration
- Select OK to reboot (but you won’t reboot yet)
- Set your time zone as required. If you select Yes, use arrow keys to move up/down and press <space> to select, then <tab> to move to the OK selection
- Select Yes (or No if you didn’t use the <space> key as directed 😊) to verify the time zone is correct
- You are now asked about the hostname. This is what the node is called on your network. We suggest you change this to <your call>-<your node number> For example, w8xyz-54321. The choice is up to you but use lower case if you change it
- Next you are asked about the IP address. Leave it DYNAMIC unless you have a reason to change it
- Next you are asked about the SSH port. Leave it as 222 unless you have reason to change it

- Next, note the IP address and port number and select REBOOT
- Wait for the blinking green HB LED and the IP address announcement then press R to restart the session
- Enter your new password (you wrote it down – right?)

Configure Asterisk:

- You are now going to reconfigure some Asterisk configuration files. Select YES
- Enter your node number
- Enter your call
- Report your status. Select YES
- Choose between voice ID and CW ID. Your choice but voice ID can become a bit obtrusive after a while
- Enter your binding port (called IAX port by Allstar). This was selected when you created a server before requesting a node number in allstarlink.org. For your first node it is 4569 as shown in the entry box. If this is not your first node, it needs to be different than your other nodes. **This means that each node number you have must have its own server and unique IAX port.**
- Choose duplex mode 1 semi-half duplex
- Enter the password for your node as shown. This is NOT the password you created to log into HamVOIP previously in this setup
- Choose NO for a password for an iaxrpt connection.
- Choose NO for Simple USB settings
- Choose YES to restart asterisk

You will now see the Admin Menu -phew!

Hint: If you mess up and can not recover, reboot by turning your node OFF and back ON. You will boot into the Admin Menu. Select Item 9 Start bash shell interface. Enter **firsttime and press enter to restart the setup.**

Set up simpleusb:

- Use the down arrow key on your keyboard to select Item12 in the Admin Menu and then <enter> to Run Selected Item
- You now see the simple-usb-tune menu
 - Select 2, then press <enter><enter>. It shows the current setting. Enter 600 <enter>. Press <enter> again and again to return to the menu selection. (The remaining selections are easier!)
 - Select 3, then enter 600 and press <enter>

- Select 7 and enter 310, then press <enter>
- Select B and then press <enter>. Verify that B in the menu has been toggled to Disabled
- Select H and then press <enter>. Verify that H in the menu has been toggled to Disabled
- Select P and you should see:

Card is: 0

Name is: usb

Rx Level: 600

Rx ondelay: 0

Rx audio-delay: 0

Tx A Level: 600

Tx B Level: 500

Tx DSP Level: 310

preemphasis: no

deemphasis: no

plfilter: no

dcsfilter: no

rxboost: no

dcsfilter: no

rxboost: no

PTT: Active LOW

PTT status: Un-Keyed

CTCSS (input): Ignored

COS (input): CM108/CM119 Active LOW

COS (test): Un-Keyed

COS (composite): Un-Keyed

- Press <enter> to continue
- **Select W to save your setup and press <enter> to continue**

- Select 0 (zero) and press <enter> to Exit the simpleusb setup and return to the Admin Menu

Set up the K4H818 radio module:

In order to setup the K4H818 radio module for parameters such as RF frequency, squelch, PL usage and frequency, filter usage, etc., we install several programs and use them to setup the radio and flash new firmware. This setup is done from the bash shell of the Arch Linux used by HamVOIP.

- Select Item 9 of the Admin Menu. You will be sent to the Bash Shell where you will see a prompt like [root@w8xyz-54321 ~]#. This is the terminal. You enter commands (type them in) and then press <enter> to execute them. Many of them are quite long and complicated so we give them to you here in bold. You can copy them with a highlight then left click and select copy (do not use <ctrl>-c). Then paste them to the terminal by using a left click to return focus to the command line and a right click to paste (do not use <ctrl>-v). Got it?
- Enter **wget -O - https://raw.githubusercontent.com/Kits4Hams/K4H818_utilities/main/install_hamvoip.sh | bash**
This installs K4H818 utility programs in the /usr/local/sbin directory and a program called AVRdude (normally already installed in HamVOIP) used to reflash the K4H818. The installation process will end with the words:

Done. Run with:

/K4H818-prog.py

/K4H818-update.py

Hint: Re-run this same installer (i.e. the above **wget**) any time to update both to the latest K4H818-prog.py and K4H818-update.py versions and re-check avrdude.

- Enter **cd /usr/local/sbin**
- Enter **ls -l K4H818*** and verify that the following programs are found
-rwxr-xr-x 1 root root 58142 Sep 1 15:16 K4H818-prog
-rwxr-xr-x 1 root root 10176 Sep 1 15:16 K4H818-update
-rw-r--r-- 1 root root 2694 Sep 1 15:16 K4H818_example.ini

Update K4H818 firmware

- Enter **K4H818-update** to update the K4H818 radio module to the latest firmware. Verify firmware update is successful.

Program the K4H818 radio module:

- Enter **K4H818-prog** to start the K4H818 programming app.

Hint: Use K4H818-prog -d to use pre-selected configurations. Use <ctrl>+c to start over

The current configuration as read from the K4H818 is presented. Then an interactive session is started. Each new question presents the current setting which can be selected by pressing <enter>. There are 5 subaudible tone types possible. We recommend using a subaudible tone (choices 1 thru 4). Squelch can be set from 0 to 8. We recommend an RX volume setting of 30. We recommend all filters be enabled (i.e. y) Here is a sample output.

Current configuration read from the K4H818:

Firmware Version: 1.5.13
Tx Frequency: 435.000 MHz
Rx Frequency: 435.000 MHz
Tx Sub-audible Type: CTCSS
Tx CTCSS Frequency: 100.0 Hz
Rx Sub-audible Type: CTCSS
Rx CTCSS Frequency: 100.0 Hz
Squelch Value: 5
RX Volume: 30
Pre/De-Emphasis Enabled: y
Voice High Pass Enabled: y
Voice Low Pass Enabled: y
CTCSS/CDCSS High Pass Enabled: y
CTCSS/CDCSS Low Pass Enabled: y
RSSI Low Pass Filter Enabled: y

Verify/Adjust RX Voice Level:

- Set up a radio to the frequency and subaudible tone you just programmed into the K4H818
- Key up and verify the yellow COS and blue CSS LED come on.
- Unkey and verify the node transmits a courtesy tone and the red PTT LED flashes on
- Key up and connect to the Parrot node (55553) by dialing DTMF sequence *355553 and unkey
- After the connection message is announced by your node, key up and ID with your call and a short voice message. Unkey and listen for the comment on level followed by a parrot of your voice message. If the level is not correct, change the distance of your mouth to the microphone on the radio, change the mic gain on your radio or go to Set Up Simple USB above and change 2) RX Voice Level = 600 to a different value. You can just keep adjusting the level from 600 until you have it correct. Dial *10 to disconnect from the parrot node

Install Supermon:

In order to install supermon, you must edit two files. We will use the file editing capability of MobaXterm to do this. This capability is in the left panel file manager capability of MobaXterm below Quick connect..... You navigate between directories by clicking on the yellow folder with up arrow/folder icon



to move up to the root and then clicking on the directories themselves to enter them.

From the Admin Menu, Item 9 Bash Shell terminal, enter or copy and paste the following to install and run *supermon_latest update* as follows:

- **cd /usr/local/sbin**
- **wget "http://2577.nodes.allstarlink.org:43856/supermon_latest_update" -O supermon_latest_update**
- **chmod +x supermon_latest_update**
- **hash**
- **supermon_latest_update**

To set the password for supermon to connect with asterisk use the file editing capability of MobaXterm

- Navigate to /etc/asterisk by returning to root and then double left clicking on the etc folder and then the asterisk folder
- Double click on the manager.conf file
- In the [admin] stanza (near the bottom), add a password to secret = (for example: secret = icn3toap4totoe). Remember this password for use later in this procedure.
- Click on File>Save>Yes Your file will save and collapse to the bottom of the screen as MobaTextEditor. Right click on it and X to close it

To configure allmon.ini for supermon, again use the file editing capability of MobaXterm

- Navigate to /srv/http/supermon by returning to the root folder and clicking on the srv folder, then http folder, then the supermon folder
- Double click on allmon.ini to edit it using the MobaXterm editor
- Change stanza [1998] to [your node number]
- Enter the password you created when you just edited manager.conf after passwd= in the first stanza
- Put semicolons in front of the [1999] stanza and all lines in the stanza (i.e. turn them into comments)
- In the [All Nodes] stanza, change 1998,1999 to your node number
- In the [lnodes] stanza change 1998 to your node number
- Click on File>Save>Yes Your file will save and collapse to the bottom of the screen. Right click it and click X to close it

To create your supermon login user and password continue at the bash shell termina (the right window in MobaXterm with the black background) Enter the following lines:

- **rm .htpasswd** (continue on if No such file or directory message)
- **htpasswd -cB .htpasswd admin** (if a different user name than admin is desired change admin to it here – we like to use our call sign)
Hint: You will be asked to enter the password twice, typing of the password does not show on the screen
- **reboot**
- Wait for the reboot blinking green LED then press R to restart the session

Verify you can log into supermon

- Open a new tab in your browser
- Enter <**your node's IP address**>/supermon as the URL
- Click on your node number (in the blue line) to activate the node number (must do this to log in)
- Click on Login (upper left corner of the screen below Your CALL.....) then login using Tab key to navigate - not the mouse

Finish Supermon Setup:

Use the MobaXterm file editor to edit the supermon screen by making changes in global.inc

- Navigate to /srv/http/supermon by returning to the root and clicking on srv, then http, then supermon
- Double click on global.inc to edit it
- You can change anything in quotes. We Recommend:

```
$CALL = "Your Callsign";  
$NAME = "Your Name";  
$LOCATION = "Your city, state";  
$TITLE2 = "RPI2-3 Node"; (we recommend a description of your node e.g. SHARI PiHatU NeoPi5)  
$TITLE3 = "AllStarLink/IRLP/EchoLink/Digital - Bridging Control Center"; (we recommend "Node <your node number> ")  
$BACKGROUND = "background.jpg"; (we recommend deleting background.jpg so you have two quotes i.e. "")  
$BACKGROUND_COLOR = "blue";  
$LOCALZIP = "39904"; (your zip code to get your local weather)
```

- Click on File>Save>Yes Your file will save and collapse to the bottom of the screen. Hover over it and click X to close it
- Refresh your browser window containing supermon to see the changes

WiFi:

Here is how to set up WiFi on the HamVOIP image

- Click on (select) Item 7 of the Admin menu and press <enter>
- Select Item 1 Setup wireless IS and passphrase
- Continue (Yes)
- Select <No> to scan for available networks
- Select the desired SSID (WiFi network)
- Enter your WiFi passphrase (password) for the network
- Select <Yes> and then carefully read and understand the displayed message. Be especially aware of the part of the message pertaining wired and wireless connections at the same time

```
If you are using a wired Ethernet connection to the server you
will need to reboot by selecting 'YES'. You will then lose
connection. You should then immediately disconnect the wired
connection to the server during the reboot. Once rebooted if all
went well you should be connected the the wireless access point
your selected.
```

```
It is important to not have both wired and wireless connections at
the same time during normal operation.
```

Shutdown:

- In MobaXterm at the Bash shell prompt enter **exit** (if necessary to return to the admin menu)
- In the Admin Menu select 16 Perform system power down